

Quadratic Functions & Equations

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **identify** key features of a quadratic function from its graph or equation.

RELATED SKILLS

☐ I Determine the vertex and axis of symmetry.☐ I Determine x- and y-intercepts of a quadratic function.☐ I Identify standard, factored, and vertex forms.☐ I Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.

☐ 2. I can **connect** standard, factored, and vertex forms to features of a parabola.

RELATED SKILLS

☐ I Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.☐ D Connect parameters in a formula to features of its graph.☐ D Identify standard, factored, and vertex forms.☐ A Determine the vertex and axis of symmetry.☐ A Determine x- and y-intercepts of a quadratic function.

☐ 3. I can **solve** a quadratic equation by factoring.

RELATED SKILLS

☐ I Determine real and complex zeros using an appropriate method.☐ D Apply the zero-product property.☐ A Factor a monic quadratic trinomial.☐ A Factor a quadratic trinomial with a nonunit leading coefficient.

☐ 4. I can **solve** a quadratic equation using the square-root property.

RELATED SKILLS

☐ I Apply the square-root property with both signs.☐ A Extract perfect-power factors from a radical.

☐ 5. I can **solve** a quadratic equation by completing the square.

RELATED SKILLS

☐ I Create a perfect-square trinomial by completing the square.☐ A Simplify an algebraic expression using operation properties.

☐ **6. I can **solve** a quadratic equation using the quadratic formula.**

RELATED SKILLS

- ☐ I Substitute coefficients accurately into the quadratic formula.
 - ☐ I Use the discriminant to predict the number and type of solutions.
 - ☐ A Extract perfect-power factors from a radical.
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☐ **7. I can **choose** an efficient method for solving a quadratic equation.**

RELATED SKILLS

- ☐ I Select an efficient quadratic-solving method from equation structure.
 - ☐ A Apply the square-root property with both signs.
 - ☐ A Apply the zero-product property.
 - ☐ A Create a perfect-square trinomial by completing the square.
 - ☐ A Substitute coefficients accurately into the quadratic formula.
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☐ **8. I can **graph** a quadratic function using its vertex, intercepts, and symmetry.**

RELATED SKILLS

- ☐ D Connect parameters in a formula to features of its graph.
 - ☐ D Determine the vertex and axis of symmetry.
 - ☐ D Determine x- and y-intercepts of a quadratic function.
 - ☐ A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
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☐ **9. I can **create and interpret** a quadratic model for a projectile, area, or optimization problem.**

RELATED SKILLS

- ☐ I Interpret vertex and zeros in an application.
 - ☐ D Explain a model parameter in the language of its context.
 - ☐ A Attach and interpret units for rates, inputs, and outputs.
 - ☐ A Restrict a model to meaningful contextual inputs.
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